

ENGINEERING FEATURE REFERENCE

Eurskem AI Complete Feature Catalog

Product surfaces, workflow capabilities, AI and RAG, integrations, security, operations, tests, and registered node types.

Agentic Workflow Platform

Catalog date: 24 August 2026

Derived from the current repository

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Repository: `agentic-workflow-platform`

Catalog date: 24 August 2026

Source basis: current React routes and components, FastAPI routers, workflow runtime, registered node types, Chat workspace presets, Knowledge Studio, integrations, tests, and deployment configuration.

This document lists the features currently represented in the Eurskem AI codebase. It separates visible product features from reusable platform capabilities. A configured external provider may still be required for a feature to operate in a particular deployment.

Status notation

- **Current** — represented by current product or backend code.
- **Configuration-dependent** — current implementation exists, but credentials or infrastructure must be configured.
- **Compatibility** — retained for saved-workflow compatibility but not preferred for new authoring.
- **Not current** — explicitly absent, retired, or deferred.

At-a-glance product surfaces

Surface	Main purpose	Primary implementation	--- --- ---	Chat	Conversational use of workflows, sources, tools, and artifacts
					<code>ui/src/modes/studio/business-chat/</code> , <code>app/api/chat_workspace.py</code> , <code>app/api/run_chat.py</code>
					Workflows
					Discover, inspect, import, generate, prepare, and launch workflows
					<code>ui/src/modes/studio/Library.tsx</code> , <code>app/api/workflows.py</code>
					Builder
					Visual and YAML workflow authoring
					<code>ui/src/modes/studio/Builder.tsx</code> , <code>app/api/builder.py</code>
					Cockpit
					Live technical workflow execution view
					<code>ui/src/modes/studio/Cockpit.tsx</code> , <code>app/runtime/</code> , <code>app/api/runs.py</code>
					Run History
					Durable run inspection, retry, diagnosis, and Ask AI
					<code>ui/src/modes/studio/RunHistory.tsx</code> , <code>app/workflow/run_history.py</code>
					Knowledge Studio
					Collection, ingestion, retrieval, profile, RAG-agent, and trace management
					<code>ui/src/modes/modes/knowledge/</code> , <code>app/api/knowledge.py</code>
					Evaluation Lab
					Golden-set scoring and model comparison
					<code>ui/src/modes/eval/EvalRoot.tsx</code> , <code>app/api/eval.py</code>
					Cost Management
					Usage, pricing, infrastructure, cache, and budget administration
					<code>ui/src/modes/cost/CostRoot.tsx</code> , <code>app/api/cost*.py</code>

1. Authentication, shell, and navigation

ID	Feature	Status	Implementation	--- --- --- ---	AUTH-01	Username/password sign-in	Current	ui/src/components/auth/LoginPage.tsx , app/api/auth.py
					AUTH-02	HttpOnly-cookie authentication	Current	app/api/auth.py , app/security/jwt_handler.py
					AUTH-03	Session rehydration after browser refresh	Current	ui/src/App.tsx , GET /auth/me
					AUTH-04	Session-expiry detection with a clear re-login notice	Current	ui/src/App.tsx , ui/src/api/client.ts
					AUTH-05	Logout and cookie invalidation	Current	POST /auth/logout
					AUTH-06	Role- and permission-based API access	Current	app/security/dependencies.py , app/security/rbac.py
					SHELL-01	Responsive application shell	Current	ui/src/App.tsx , layout components
					SHELL-02	Collapsible desktop sidebar	Current	ui/src/App.tsx , Sidebar.tsx
					SHELL-03	Mobile navigation drawer	Current	ui/src/App.tsx , Sidebar.tsx
					SHELL-04	Lazy-loaded top-level modes	Current	ui/src/App.tsx
					SHELL-05	Global run-cost display in the top bar	Current	RunCostContext , Topbar.tsx
					SHELL-06	Four top-level modes: Studio, Knowledge, Evaluation, Cost	Current	ui/src/App.tsx

2. Workflow-powered Chat

| ID | Feature | Status | Implementation | |---|---|---|---| CHAT-01 | Source-first conversational workspace | Current | [SourceFirstChatHome.tsx](#), [BusinessChat.tsx](#) | CHAT-02 | General AI conversation through managed workflow adapters | Current | [chat_workspace_planner.py](#) | CHAT-03 | Execute an existing saved workflow from Chat | Current | [build_existing_workflow_adapter\(\)](#) | CHAT-04 | Persist private Chat workflows | Current | [app/workflow/chat_workflow_store.py](#) | CHAT-05 | Shared and private workflow conversations | Current | [chat_conversation_store.py](#), Chat routes | CHAT-06 | Conversation/session history panel | Current | [ChatSessionsPanel.tsx](#) | CHAT-07 | Follow-up turns in the same workflow session | Current | [BusinessChat.tsx](#), run-chat APIs | CHAT-08 | Follow-up turns retain expanded document context | Current | [app/api/run_chat.py](#), file adapters | CHAT-09 | Unified Add Sources hub | Current | [ChatWorkspaceOverlays.tsx](#), [ComposerMenu.tsx](#) | CHAT-10 | Local file uploads as Chat sources | Current | workflow-input-file APIs | CHAT-11 | Pasted images uploaded as sources | Current | Chat composer/source model | CHAT-12 | Web URLs as explicit sources | Current | Chat source model and planner | CHAT-13 | Google Drive OAuth and account browsing | Configuration-dependent | [app/integrations/files/google_drive.py](#), [integration_oauth.py](#) | CHAT-14 | Google Drive search, multi-select, and secure import | Configuration-dependent | [CloudFileBrowser.tsx](#), file integration service | CHAT-15 | OneDrive integration framework | Configuration-dependent | [app/integrations/files/onedrive.py](#) | CHAT-16 | Imported cloud-file provenance | Current | file integration metadata and source model | CHAT-17 | Knowledge collection as a Chat source | Current | [NotebookSourcesPanel.tsx](#), planner | CHAT-18 | Individual Knowledge-document selection | Current | [document_ids](#) planning and retrieval scope | CHAT-19 | Add and remove sources before or during a conversation | Current | source panels and workspace model | CHAT-20 | Source state restoration on desktop and mobile | Current | [chatWorkspaceStorage.ts](#) | CHAT-21 | Automatic routing among LLM, files, vision, web, retrieval, integration, workflow, and artifact plans | Current | [plan_workspace\(\)](#) | CHAT-22 | Web-search route for current/public information | Current | web adapter and [WebSearchAgent](#) | CHAT-23 | Image and mixed-document analysis | Current | vision and multimodal file adapters | CHAT-24 | Image, diagram, infographic, PDF, DOCX, and PPTX output routing | Current | artifact adapters | CHAT-25 | Scientific skill selection and execution | Current | [ScientificSkillAgent](#), research skills API | CHAT-26 | Prompt-template library | Current | [PromptTemplateLibrary.tsx](#), prompt-template APIs | CHAT-27 | Speech input support | Browser-dependent | [speech.ts](#) | CHAT-28 | Live agent activity groups and step cards | Current | [AgentActivityGroup.tsx](#), [AgentActivityCard.tsx](#) | CHAT-29 | Chat node inspector | Current | [ChatNodeInspector.tsx](#) | CHAT-30 | Human intervention/approval cards inside Chat | Current | [ChatInterventionCard.tsx](#) | CHAT-31 | Run controls from Chat | Current | [RunControlBar.tsx](#), run-control model | CHAT-32 | Workflow context panel | Current | [WorkflowContextPanel.tsx](#) | CHAT-33 | Session audit panel | Current | [SessionAuditPanel.tsx](#) | CHAT-34 | Downloadable structured and file outputs | Current | [chatOutputs.ts](#), artifact download routes | CHAT-35 | Retry malformed empty model answers such as `null`, `[]`, and `{}` | Current | [app/api/run_chat.py](#) | CHAT-36 | User-facing fallback when a valid answer still cannot be extracted | Current | [chatOutputs.ts](#) | CHAT-37 | Retrieval-timeout retry with checkpoint-backed recovery | Current | runtime/retrieval retry paths | CHAT-38 | Managed adapter caching by semantic source scope | Current | [app/api/chat_workspace.py](#) |

2.1 Chat experience presets

The planner currently exposes these 20 guided experiences:

1. Ask Questions About My Documents
2. Research Analyst
3. Research to Presentation
4. Research to PDF Report
5. Meeting / Interview Intelligence
6. Customer Feedback Analysis
7. Competitive Intelligence
8. Contract / Policy Understanding
9. Long Document Assistant
10. Study / Learning Assistant
11. Executive Brief Generator
12. Data / Results Interpreter
13. Product Requirements Assistant

14. Content Repurposing
15. Proposal Generator
16. Due-Diligence Assistant
17. Incident / Troubleshooting Assistant
18. Decision Support
19. Chat → Workflow Execution
20. Multi-Workflow AI Project

Source: [app/workflow/chat_workspace_planner.py](#) .

3. Workflow Library

ID	Feature	Status	Implementation	
LIB-01	Browse canonical workflows by outcome	Current	Library.tsx	
LIB-02	Search workflows	Current	library toolbar/filter utilities	
LIB-03	Category navigation and category counts	Current	CategoryNav.tsx , categories.ts	
LIB-04	Filter by workflow characteristics and outputs	Current	filters.ts , LibraryToolbar.tsx	
LIB-05	Sort workflows	Current	library filters/sort model	
LIB-06	Grid and list presentation	Current	LibraryToolbar.tsx	
LIB-07	Favorite workflows locally	Current	library/localState.ts	
LIB-08	Recently opened and “continue where you left off”	Current	library/localState.ts	
LIB-09	Workflow detail panel	Current	WorkflowDetailsPanel.tsx	
LIB-10	Overview, requirements, outputs, stages, evidence, versions, runs, and technical detail tabs	Current	library/tabs/	
LIB-11	Readiness and preflight summary	Current	library/readiness.ts	
LIB-12	Prepare workflow inputs before launch	Current	PrepareAndRunPanel.tsx	
LIB-13	Launch a workflow into Cockpit	Current	workflow run APIs	
LIB-14	Import workflow YAML	Current	ImportWorkflowDialog.tsx	
LIB-15	Generate workflow YAML from a prompt	Current	GenerateWorkflowDialog.tsx , workflow-generation API	
LIB-16	Create a blank workflow	Current	Builder navigation	
LIB-17	Delete eligible workflows with confirmation	Current	ConfirmDeleteDialog.tsx	
LIB-18	Refresh workflow catalog	Current	Library.tsx	

4. Visual Workflow Builder

| ID | Feature | Status | Implementation | |---|---|---|---| BUILD-01 | Visual graph editing with React Flow | Current | [Builder.tsx](#) | | BUILD-02 | Registry-driven node palette | Current | [NodePalette.tsx](#) , node-type API | | BUILD-03 | Searchable node/task palette | Current | [NodeSearchPalette.tsx](#) | | BUILD-04 | Drag/drop node creation and edge editing | Current | [Builder.tsx](#) | | BUILD-05 | Typed schema-generated configuration forms | Current | [SchemaForm.tsx](#) | | BUILD-06 | Purpose-built editors for specialized nodes | Current | [ConfigPanel.tsx](#) , Builder inspector components | | BUILD-07 | Workflow input editor | Current | [WorkflowInputsPanel.tsx](#) | | BUILD-08 | Workflow variable editor | Current | [WorkflowVariablesPanel.tsx](#) | | BUILD-09 | Visual input/output data mapping | Current | [DataMappingPanel.tsx](#) | | BUILD-10 | YAML-to-canvas and canvas-to-YAML round trip | Current | [yaml-bridge.ts](#) | | BUILD-11 | Direct YAML editing and validation | Current | Builder/YAML bridge | | BUILD-12 | Zero-token preflight panel | Current | [PreflightPanel.tsx](#) , preflight API | | BUILD-13 | Workflow autofix | Current | Builder autofix APIs | | BUILD-14 | Node-level autofix | Current | Builder node-autofix path | | BUILD-15 | Single-node test execution | Current | [BuilderTestPanel.tsx](#) , [/api/builder/node-test](#) | | BUILD-16 | Branch/graph simulation | Current | [/api/builder/simulate](#) | | BUILD-17 | Schema preview and output-contract tools | Current | Builder API | | BUILD-18 | AI-assisted schema drafting | Current | [/api/builder/assist/schema](#) | | BUILD-19 | AI-assisted rule drafting | Current | [/api/builder/assist/rules](#) | | BUILD-20 | Prompt-drafting assistant | Current | [PromptDraftAssistant.tsx](#) | | BUILD-21 | Ask AI about a node type | Current | [NodeTypeAskAi.tsx](#) , node-types Chat API | | BUILD-22 | LLM model selector and routing policy | Current | [ModelSelect.tsx](#) | | BUILD-23 | MCP server/tool discovery | Configuration-dependent | Builder MCP routes | | BUILD-24 | MCP tool health and test execution | Configuration-dependent | Builder MCP routes | | BUILD-25 | Email connection discovery and OAuth | Configuration-dependent | Builder email APIs | | BUILD-26 | File/cloud integration discovery and OAuth | Configuration-dependent | integration OAuth APIs | | BUILD-27 | Autosave serialized to prevent stale writes | Current | [Builder.tsx](#) | | BUILD-28 | Recover autosaved drafts | Current | Builder draft store | | BUILD-29 | Immutable workflow versions and version history | Current | [VersionHistoryPanel.tsx](#) | | BUILD-30 | Save As workflow | Current | [SaveAsDialog.tsx](#) | | BUILD-31 | Undo/redo | Current | Builder snapshots | | BUILD-32 | Add non-executable notes to the canvas | Current | [NoteNode.tsx](#) | | BUILD-33 | Automatic graph layout | Current | [flow-layout.ts](#) | | BUILD-34 | Left-to-right and top-to-bottom layouts | Current | Builder view state | | BUILD-35 | Stage grouping, bands, collapse, and placeholders | Current | [builder/stage-view.ts](#) | | BUILD-36 | Semantic zoom and compact node rendering | Current | Builder zoom tiers | | BUILD-37 | Fullscreen/expanded canvas | Current | [Builder.tsx](#) | | BUILD-38 | Minimap, controls, and graph navigation | Current | React Flow and navigation utilities | | BUILD-39 | Export workflow graph to SVG | Current | [graph-export.ts](#) | | BUILD-40 | Export workflow graph to PNG | Current | [graph-export.ts](#) | | BUILD-41 | Launch current draft for execution | Current | [RunDialog.tsx](#) | | BUILD-42 | Reference pruning and rename safety when graph nodes change | Current | [builder-graph.ts](#) |

5. Workflow runtime and execution

ID	Feature	Status	Implementation	Progress
RUN-01	Typed YAML workflow schema	Current	app/runtime/schema.py	
RUN-02	LangGraph compilation	Current	app/runtime/compiler.py	
RUN-03	Zero-token schema and preflight validation	Current	app/runtime/preflight.py	
RUN-04	Graph topology validation	Current	runtime preflight modules	
RUN-05	Template/reference validation	Current	templating.py , logic_preflight.py	
RUN-06	Typed workflow inputs and variables	Current	runtime schema	
RUN-07	Typed node configuration, input, and output schemas	Current	NodeType contract	
RUN-08	Runtime output validation after every node	Current	node/runtime boundary	
RUN-09	Sequential execution	Current	graph edges	
RUN-10	Parallel fan-out through graph edges	Current	compiler/runtime	
RUN-11	Conditional branch routing	Current	RouterAgent , conditional edges	
RUN-12	Deterministic business-rule decisions	Current	DecisionAgent	
RUN-13	Human-in-the-loop pause, approve, reject, and edit	Current	HumanInLoopAgent , HITL runtime	
RUN-14	Durable resume after human intervention	Current	app/runtime/hitl.py	
RUN-15	Cooperative pause requests	Current	runtime/run store	
RUN-16	Run cancellation handling	Current	run APIs/runtime	
RUN-17	Checkpoint-backed retry of failed runs	Current	retry checkpoint APIs	
RUN-18	Reuse successful upstream node outputs during retry	Current	checkpoint runtime	
RUN-19	Child/subworkflow execution	Current	SubprocessAgent	
RUN-20	Durable run records	Current	app/workflow/run_history.py	
RUN-21	Node lifecycle timing and status	Current	runtime and run history	
RUN-22	Audit records	Current	audit APIs/store	
RUN-23	Per-node cost context	Current	LLM gateway/cost ledger	
RUN-24	Background run ownership and orchestration	Current	BackgroundRunManager	
RUN-25	Redis ownership leases	Configuration-dependent	app/runtime/coordination.py	
RUN-26	LangGraph Redis checkpoints with durable fallback	Configuration-dependent	app lifespan/HITL	
RUN-27	Idempotent external-operation ledger	Current	app/integrations/operations.py	

6. Cockpit and live execution monitoring

| ID | Feature | Status | Implementation | |---|---|---|---| | LIVE-01 | Live workflow graph | Current | [Cockpit.tsx](#) | | LIVE-02 | Node statuses: pending, running, paused, completed, rejected, failed, skipped/cancelled | Current | Cockpit state model | | LIVE-03 | Server-Sent Events progress stream | Current | [RunEventBus](#), run events API | | LIVE-04 | Redis-backed event replay across workers | Configuration-dependent | [app/runtime/events.py](#) | | LIVE-05 | Manual authenticated stream client | Current | [ui/src/api/client.ts](#) | | LIVE-06 | Auto-follow active node | Current | [Cockpit.tsx](#) | | LIVE-07 | Active execution-path highlighting | Current | [cockpit-state.ts](#) | | LIVE-08 | Stage grouping and collapse | Current | Cockpit graph-collapse modules | | LIVE-09 | Show only active path | Current | [Cockpit.tsx](#) | | LIVE-10 | Fullscreen graph and output views | Current | [Cockpit.tsx](#) | | LIVE-11 | Resizable overview and inspector panels | Current | [useResizablePanel.ts](#) | | LIVE-12 | Node overview, input, output, errors, logs, and technical inspection | Current | [NodeInspector.tsx](#), Cockpit tabs | | LIVE-13 | Human-review panel | Current | [HITLPanel.tsx](#) | | LIVE-14 | Run output viewer and artifact download | Current | [OutputViewer.tsx](#) | | LIVE-15 | Live run cost summary | Current | Cockpit hook and cost API | | LIVE-16 | Stream failure and reconnect handling | Current | [useCockpitRun.ts](#) |

7. Run History and diagnosis

ID Feature Status Implementation --- --- --- ---	HIST-01 Durable run list and run selection Current
RunListPanel.tsx	HIST-02 Run overview Current OverviewTab.tsx
HIST-03 Node-by-node history Current	NodesTab.tsx
HIST-04 Inputs view Current InputsTab.tsx	HIST-05 Outputs and artifact view Current
OutputsTab.tsx	HIST-06 Timeline and audit view Current TimelineTab.tsx
HIST-07 Error diagnosis view Current	ErrorsTab.tsx
HIST-08 Ask AI about a completed/failed run Current AskAiPanel.tsx , run-chat API	HIST-09
Historical node inspector Current Cockpit NodeInspector reused in Run History	HIST-10 Retry failed runs from
checkpoints Current Run History data hook and retry API	HIST-11 Autofix failed workflow and reopen Builder Current
Run History data hook	HIST-12 Open a run in Cockpit Current Run History navigation
HIST-13 Open proposal	review from a run Current Run History navigation
HIST-14 Open evidence candidates from a run Current Run History	navigation
HIST-15 Resizable and fullscreen inspection Current Run History panels	

8. Knowledge Studio, ingestion, and retrieval

| ID | Feature | Status | Implementation | |---|---|---|---| | KNOW-01 | Create and manage Knowledge collections | Current | Collections page/API | | KNOW-02 | Collection-scoped UI context | Current | [CollectionContext.tsx](#) | | KNOW-03 | Document upload and ingestion | Current | [IngestionPage.tsx](#), ingestion APIs | | KNOW-04 | Parser profiles | Current | Knowledge models/strategies | | KNOW-05 | Standard, layout-aware, structure-aware, OCR-fallback, and vision-augmented parsing strategies | Current | [app/knowledge/models.py](#) | | KNOW-06 | Chunking profiles and strategies | Current | Knowledge models/chunker | | KNOW-07 | Embedding profiles | Current | Knowledge models/embedders | | KNOW-08 | Document and index inventory | Current | [DocumentsIndexesPage.tsx](#) | | KNOW-09 | Index creation and management | Current | Knowledge APIs/services | | KNOW-10 | Retrieval Playground | Current | [PlaygroundPage.tsx](#) | | KNOW-11 | Dense retrieval | Current | retrieval profile strategy | | KNOW-12 | Sparse retrieval | Current | retrieval profile strategy | | KNOW-13 | Hybrid retrieval | Current | retrieval service | | KNOW-14 | Hybrid retrieval with reranking | Current | retrieval service/reranker | | KNOW-15 | Retrieval-result comparison | Current | [/api/retrieval/compare](#) | | KNOW-16 | Metadata filter validation and compilation | Current | [app/retrieval/filters.py](#) | | KNOW-17 | Mandatory tenant/session isolation | Current | retrieval filters | | KNOW-18 | Collection security scope | Current | retrieval models/filter compiler | | KNOW-19 | Server-resolved document-level security scope | Current | [document_ids](#) retrieval path | | KNOW-20 | Date, type, industry, and collection filters | Current | retrieval models | | KNOW-21 | Self-query metadata filtering without security-scope override | Current | retrieval filter compiler | | KNOW-22 | Retrieval routing profiles: deterministic, AI, and hybrid | Current | Knowledge models | | KNOW-23 | Generation profiles | Current | Knowledge models | | KNOW-24 | Retrieval Profiles management | Current | [ProfilesAgentsPage.tsx](#) | | KNOW-25 | RAG Agent creation, listing, inspection, and querying | Current | RAG Agent APIs | | KNOW-26 | Retrieval traces | Current | [TracesPage.tsx](#), trace APIs | | KNOW-27 | Open a retrieval trace from another product surface | Current | browser events/local storage handoff | | KNOW-28 | Retrieval candidates, rerank reasons, latency, and provenance | Current | retrieval result contracts | | KNOW-29 | Citation-grounded generation | Current | RAG service and nodes | | KNOW-30 | Knowledge retrieval without answer generation | Current | [KnowledgeRetrieval](#) node | | KNOW-31 | Retrieval cache identity scoped by selected documents | Current | RAG/Chat adapter cache identity |

9. AI, LLM, and model routing

ID	Feature	Status	Implementation						
AI-01	Multi-provider LLM gateway	Configuration-dependent	app/llm/						
AI-02	OpenAI provider support	Configuration-dependent	OpenAI SDK/client						
AI-03	Anthropic provider support	Configuration-dependent	Anthropic SDK/client						
AI-04	OpenRouter provider support	Configuration-dependent	registry/OpenRouter modules						
AI-05	Local-model capability probes	Configuration-dependent	app lifespan/LLM registry						
AI-06	Model registry and aliases	Current	app/llm/registry.py						
AI-07	Capability-aware model selection	Current	model router						
AI-08	Automatic model routing	Current	model router/OpenRouter ranking						
AI-09	Accuracy-priority routing policies	Current	model routing policy						
AI-10	Retry with provider Retry-After handling	Current	LLM gateway						
AI-11	Ordered fallback chains	Current	LLM gateway/router						
AI-12	Intended-versus-actual model audit	Current	cost ledger						
AI-13	Structured-output generation and validation	Current	AI/Transform nodes						
AI-14	Configurable AI extraction, classification, summarization, translation, rewriting, generation, and analysis	Current	AI/Transform modes						
AI-15	Prompt caching metrics and management	Provider-dependent	Cost Management cache tab						
AI-16	LLM policy evaluation	Configuration-dependent	app/security/llm_policy.py						
AI-17	Fail-closed policy behavior	Current	security policy code						
AI-18	Kimi vision support	Configuration-dependent	KimiVisionAgent						
AI-19	OpenAI image generation	Configuration-dependent	OpenAIImageGenerationAgent						
AI-20	General image-generation service abstraction	Configuration-dependent	image tool/service						

10. Search, research, and scientific capabilities

| ID | Feature | Status | Implementation | |---|---|---|---| | RES-01 | Web search | Configuration-dependent | [WebSearchAgent](#) , web service | | RES-02 | Scholarly candidate discovery | Configuration-dependent | scholarly discovery node/MCP | | RES-03 | Bounded multi-query deep research | Configuration-dependent | [BoundedDeepResearchAgent](#) | | RES-04 | Scientific research planning | Current | [ScientificResearchPlannerAgent](#) | | RES-05 | Approved scientific skill catalog | Current | research skills API | | RES-06 | Scientific skill execution | Current | [ScientificSkillAgent](#) | | RES-07 | PaperQA evidence synthesis | Configuration-dependent | [PaperQAEvidenceSynthesizerAgent](#) | | RES-08 | Prior-project retrieval from official project records | Configuration-dependent | [PriorProjectRetrieverAgent](#) | | RES-09 | Internal project evidence retrieval | Configuration-dependent | [InternalProjectEvidenceRetrieverAgent](#) | | RES-10 | Structured dataset retrieval | Configuration-dependent | [StructuredDatasetRetrieverAgent](#) | | RES-11 | Research source acquisition and immutable storage | Configuration-dependent | [ResearchSourceAcquirer](#) | | RES-12 | Citation registry construction | Current | [CitationRegistryBuilder](#) | | RES-13 | Claim-to-source evidence verification | Current | [ClaimEvidenceVerifier](#) | | RES-14 | Evidence candidate browser | Current | [RunCandidates.tsx](#) , candidates API | | RES-15 | Verify an individual claim from candidate evidence | Current | candidates API |

11. Integrations, MCP, email, files, and external actions

ID	Feature	Status	Implementation	
INT-01	MCP server registry	Configuration-dependent	app/mcp/registry.py , connections.py	
INT-02	Dynamic MCP tool discovery and JSON schemas	Configuration-dependent	MCP service and Builder	
INT-03	Direct call of one MCP tool	Configuration-dependent	MCPToolAgent	
INT-04	LLM-driven MCP tool loop	Configuration-dependent	MCPAgent	
INT-05	Read/write policy classification for MCP tools	Current	app/mcp/policy.py	
INT-06	MCP health checks	Configuration-dependent	Builder MCP API	
INT-07	Microsoft Dynamics 365 CRM/Dataverse connector	Configuration-dependent	app/mcp/dynamics/	
INT-08	Dynamics mock/demo backend	Current	Dynamics MCP configuration	
INT-09	Dynamics 365 Finance & Supply Chain connector	Configuration-dependent	app/mcp/d365_finance/ and MCP server	
INT-10	Business-records MySQL connector	Configuration-dependent	app/mcp/business_records/	
INT-11	Guarded read-only SQL helpers	Configuration-dependent	business-record SQL guard	
INT-12	Email service abstraction	Configuration-dependent	app/integrations/email/	
INT-13	Gmail OAuth/email adapter	Configuration-dependent	gmail.py , OAuth service	
INT-14	Microsoft Graph email adapter	Configuration-dependent	msgraph.py	
INT-15	In-memory email adapter for testing/demo	Current	memory.py	
INT-16	Email connection management	Configuration-dependent	connection store and Builder APIs	
INT-17	Secure OAuth token vault for email	Configuration-dependent	email token vault	
INT-18	Google Drive file connector	Configuration-dependent	files integration package	
INT-19	OneDrive file connector	Configuration-dependent	files integration package	
INT-20	Secure OAuth token vault for cloud files	Configuration-dependent	files token vault	
INT-21	Generic integration node	Configuration-dependent	IntegrationAgent	
INT-22	Generic external REST/webhook action	Current, policy-dependent	ExternalActionAgent	
INT-23	URL validation/SSRF guard for external calls	Current	app/integrations/url_guard.py	
INT-24	Idempotent write-operation handling	Current	operation ledger	
INT-25	Database lookup tool	Configuration-dependent	database lookup service/node	

12. File handling and artifact generation

ID Feature Status Implementation	--- --- --- ---	FILE-01 Protected workflow input uploads Current workflow input file APIs/MinIO
FILE-02 Workflow-scoped file references Current	WorkflowFileRef contracts	FILE-03 PDF text extraction Current
PDFTextExtractor	FILE-04 PDF, Office, Markdown, text, and code document loading Current	WorkflowFileLoader
FILE-05 Image references retained during document loading Current	Workflow file loader	FILE-06 Styled PDF generation Current
PDFProposalRenderer	FILE-07 Styled DOCX generation Current	DOCXProposalRenderer
FILE-08 Horizon-specific DOCX generation Current	HorizonDOCXProposalRenderer	FILE-09 Horizon-specific HTML/PDF proposal generation Current
Horizon HTML renderer	FILE-10 PowerPoint generation Current	PowerPointProposalSlides
FILE-11 Excel workbook generation Current	ExcelTool	FILE-12 Dynamic figure generation Configuration-dependent
DynamicFigureAgent	FILE-13 Figure embedding into outputs Current	FigureEmbedder
FILE-14 OpenAI-generated image artifact storage Configuration-dependent	image-generation node	FILE-15 Artifact metadata and authenticated downloads Current
run/output APIs	FILE-16 Graph image export from Builder Current	SVG/PNG export

13. Proposal engineering and evidence integrity

ID Feature Status Implementation	--- --- --- ---	PROP-01 Proposal workspace and truth graph Current app/proposal_graph/
PROP-02 Proposal source/version registry Current proposal APIs/store	PROP-03 Immutable evidence versions Current source-version APIs/storage	PROP-04 Concept alternatives generation and judging Current ConceptAlternativesAgent
PROP-05 Human-approved concept freeze Current ConceptFreezeAgent	PROP-06 Call requirement coverage matrix Current CallCoverageMatrixAgent	PROP-07 Methodology engineering and method cards Current MethodologyEngineeringAgent
PROP-08 Proposal truth-graph normalization Current GraphNormalizer	PROP-09 Proposal evidence factory Current ProposalEvidenceFactoryAgent	PROP-10 Claim verification against exact passages Current evidence verifier/factory
PROP-11 Cross-document consistency checking Current ConsistencyChecker	PROP-12 Horizon proposal evaluation Current HorizonEvaluationAgent	PROP-13 Submission-readiness gate Current ProposalSubmissionGate
PROP-14 Human approval requests and decisions Current proposal approval APIs	PROP-15 Proposal Review UI Current ProposalReview.tsx	PROP-16 Render proposal to PDF Current proposal render API
PROP-17 Render proposal to DOCX Current proposal DOCX API	PROP-18 Render proposal slides Current PowerPoint node/workflows	

14. Evaluation Lab

ID Feature Status Implementation	--- --- --- ---	EVAL-01 Golden-set loading Current evaluation API/UI	EVAL-02 LLM-as-a-Judge evaluation Configuration-dependent evaluation service
EVAL-03 Faithfulness scoring Current evaluation criteria	EVAL-04 Relevance scoring Current evaluation criteria	EVAL-05 Completeness scoring Current evaluation criteria	EVAL-06 Citation-accuracy scoring Current evaluation criteria
EVAL-07 Scorecards and overall mean Current Evaluation Lab UI	EVAL-08 Evaluation history Current evaluation API/store	EVAL-09 Workflow model comparison Configuration-dependent compare API/UI	EVAL-10 Per-model pass rate, cost, and latency Current comparison results
EVAL-11 Per-case failures and checks Current Evaluation Lab UI	EVAL-12 Recommended model selection Current comparison UI		

15. Cost Management

ID Feature Status Implementation	--- --- --- ---	COST-01 Per-run cost lookup Current /api/cost/run/{run_id}	
COST-02 Cost overview for a selected date range Current Cost Overview tab		COST-03 Daily cost trend Current Cost Overview tab	
COST-04 Cost breakdown by model/provider/workflow/node Current cost admin API		COST-05 Intended-versus-actual model cost attribution Current cost ledger	
COST-06 Direct/local model pricing catalog Current Pricing tab		COST-07 OpenRouter pricing catalog Configuration-dependent pricing sync/API	
COST-08 Editable pricing overrides Admin-only Cost Pricing tab		COST-09 Revert pricing overrides Admin-only Cost Pricing tab	
COST-10 Private infrastructure cost allocation Admin-only Private Infra tab		COST-11 Prompt-cache summary Provider-dependent Prompt Cache tab	
COST-12 Budget creation and management Admin-only Budgets tab		COST-13 Budget utilization/limits Current cost admin API	

16. Security and data protection

ID Feature Status Implementation --- --- --- ---	SEC-01 Strict Pydantic request and workflow validation Current
API/runtime schemas	SEC-02 Resource ownership checks Current API routes/stores
SEC-03 Tenant/session-scoped run visibility Current run APIs/store	SEC-04 Tenant/session-scoped retrieval Current retrieval filter compiler
SEC-05 Document-level retrieval authorization Current server-resolved <code>document_ids</code>	SEC-06 Request size limiting Current security middleware
SEC-07 Trusted-host middleware Current <code>app/main.py</code>	SEC-08 Configurable CORS Current <code>app/main.py</code>
SEC-09 Request correlation IDs Current request-context middleware	SEC-10 Security response headers Current request-context middleware
SEC-11 Redis-backed rate limits Configuration-dependent rate-limit middleware	SEC-12 Production fail-closed rate limiting Current middleware policy
SEC-13 Password hashing with Argon2 Current <code>app/security/passwords.py</code>	SEC-14 JWT/token handling Current security tokens
SEC-15 Entity/PII tokenization before external model use Configuration-dependent <code>EntityTokenizerService</code>	SEC-16 OAuth state validation Current email/file OAuth APIs
SEC-17 Encrypted/secured integration token vaults Configuration-dependent integration token vaults	SEC-18 External URL and SSRF controls Current URL guard
SEC-19 MCP write policy controls Current MCP policy	SEC-20 Audit session retrieval Current audit API

17. Observability, health, and operations

ID	Feature	Status	Implementation						
OPS-01	Structured logging	Current	app/observability/logging.py						
OPS-02	Request, run, session, and node correlation	Current	logging/runtime context						
OPS-03	Prometheus metrics endpoint	Configuration-dependent	/metrics						
OPS-04	Liveness endpoint	Current	/health						
OPS-05	Dependency readiness endpoint	Current	/ready						
OPS-06	Per-service readiness latency	Current	health API						
OPS-07	MongoDB readiness	Configuration-dependent	health probes						
OPS-08	Redis readiness	Configuration-dependent	health probes						
OPS-09	MinIO readiness	Configuration-dependent	health probes						
OPS-10	Weaviate readiness	Configuration-dependent	health probes						
OPS-11	MCP readiness	Configuration-dependent	health probes						
OPS-12	Checkpointeer readiness	Configuration-dependent	health probes						
OPS-13	Prometheus alert rules for availability, errors, latency, workflow failures, failovers, and rate limits	Current	observability/prometheus/alerts.yml						
OPS-14	Docker Compose development stack	Current	docker-compose.yml						
OPS-15	Production Compose stack	Current	docker-compose.production.yml						
OPS-16	Caddy TLS and production edge controls	Current	deployment configuration						
OPS-17	Immutable release deployment	Current	deploy/ionos/deploy_release.sh						
OPS-18	Readiness/smoke-gated deployment	Current	deploy script						
OPS-19	Automatic rollback on failed readiness	Current	deploy script						
OPS-20	Mongo/MinIO/Weaviate/Redis/workflow backup	Current	deploy/ionos/backup.sh						
OPS-21	Checksummed restore	Current	deploy/ionos/restore.sh						
OPS-22	Retired workflow-artifact cleanup	Current	cleanup deployment script						
OPS-23	Database migrations	Current	app/db/migrations.py						
OPS-24	Lease-locked, idempotent migration execution	Current	migration runtime						
OPS-25	CI backend/frontend/security/build gates	Current	.github/workflows/						
OPS-26	Live-provider smoke-test workflow	Configuration-dependent	live LLM workflow/scripts						

18. Testing and quality features

ID Feature Status Implementation --- --- --- ---	QA-01 Backend unit and integration tests with Pytest Current
tests/	QA-02 Frontend unit/component tests with Vitest and Testing Library Current ui/src/**/*.test.*
QA-03	Cross-browser Playwright tests Current ui/e2e/
QA-04	Desktop, laptop, tablet, and mobile viewport matrices
Current	Playwright configuration/specs
QA-05	Accessibility testing with axe-core Current
Playwright dependency/	tests
QA-06	Workflow preflight of repository workflows without model tokens Current
preflight scripts/tests	QA-07
Node registry/audit tests	Current
node audit tests	QA-08
Builder YAML round-trip tests	Current
yaml-bridge.test.ts	QA-09
Security isolation tests	Current
security/retrieval/run tests	QA-10
Deployment and smoke	tests
Current	deployment/smoke test modules
QA-11	Generated workflow QA matrix Current
scripts/build_workflow_qa_matrix.py	QA-12
Generated browser screenshots and release-gate reports	Current artifacts
qa-results/	

19. Current registered workflow node capabilities

The live node appendix below is generated from the current `NodeRegistry`-compatible source inventory. Each node is a reusable workflow capability; compatibility-only types may be hidden from new creation while remaining executable for saved workflows.

1. **AITaskAgent** — Deprecated — use TransformAgent's Inputs/Instructions/Outputs editor instead. Category: `Uncategorized`. Source: `app/nodes/ai_task.py:424`.
2. **BoundedDeepResearchAgent** — Run multiple K-Dense-guided bounded research dossiers using a web-search tool-calling loop, with hard job, concurrency, and tool-call limits. Category: `Uncategorized`. Source: `app/nodes/bounded_deep_research_agent.py:117`.
3. **CallCoverageMatrixAgent** — Build a deterministic requirement-by-requirement Horizon call coverage matrix and submission gate. Category: `Uncategorized`. Source: `app/nodes/call_coverage.py:30`.
4. **CitationRegistryBuilder** — Deterministically reshape acquired full-text documents into a numbered, renderer-ready citation registry with canonical URLs folded into each formatted citation. Category: `Uncategorized`. Source: `app/nodes/citation_registry_builder.py:134`.
5. **ClaimEvidenceVerifier** — Verify each linked claim against an exact passage in an immutable source version. Category: `Uncategorized`. Source: `app/nodes/claim_evidence_verifier.py:62`.
6. **ConceptAlternativesAgent** — Generate conservative, balanced, and ambitious Horizon concepts grounded in the approved proposal graph. Category: `Uncategorized`. Source: `app/nodes/concept_alternatives.py:53`.
7. **ConceptFreezeAgent** — Resolve the human gate's concept decision against the three generated alternatives. Category: `Uncategorized`. Source: `app/nodes/concept_freeze.py:62`.
8. **ConsistencyChecker** — Deterministic gate. Category: `Uncategorized`. Source: `app/nodes/consistency_checker.py:149`.
9. **DataTransformAgent** — Deprecated — use TransformAgent's mode: deterministic instead. Category: `Uncategorized`. Source: `app/nodes/data_transform.py:173`.
10. **DecisionAgent** — Deterministic business rules: IF/THEN over typed fields, with nested AND/OR/NOT. Category: `Uncategorized`. Source: `app/nodes/decision.py:179`.
11. **DOCXProposalRenderer** — Render proposal sections to a styled .docx (corporate, professional, warm). Category: `Uncategorized`. Source: `app/nodes/docx_renderer.py:120`.
12. **DynamicFigureAgent** — Generate a diagram image for every `[[IMAGE PROMPT: ...]]` marker and embed it as a data URI the DOCX renderer can render; also emit a caption-only variant for text documents. Category: `Uncategorized`. Source: `app/nodes/dynamic_figure_agent.py:127`.
13. **Echo** — Renders a template string. Category: `Uncategorized`. Source: `app/nodes/_stubs.py:70`.
14. **EmailAgent** — Email in one capability: search, read, draft, reply or send, over Gmail or Microsoft — provider differences live in adapters. Category: `Uncategorized`. Source: `app/nodes/email_integration.py:229`.
15. **EndAgent** — What the workflow returns or shows when it finishes. Category: `Uncategorized`. Source: `app/nodes/end.py:109`.
16. **ExcelTableExtractor** — Extract tables from an .xlsx in object storage. Category: `Uncategorized`. Source: `app/nodes/excel_tool.py:44`.
17. **ExternalActionAgent** — Call an external REST API or send a webhook. Category: `Uncategorized`. Source: `app/nodes/external_action.py:138`.
18. **FigureEmbedder** — Replace `[[IMAGE PROMPT: marker]]` placeholders with real embedded images read from object storage, before the proposal is rendered. Category: `Uncategorized`. Source: `app/nodes/figure_embedder.py:110`.
19. **GraphNormalizer** — LLM extracts structure from source text; pydantic validates it; valid objects are written to `proposal_graph`. Category: `Uncategorized`. Source: `app/nodes/graph_normalizer.py:211`.
20. **HorizonDOCXProposalRenderer** — Convert proposal HTML or Markdown into an editable, citation-aware Horizon Europe Part B DOCX with native headings, TOC, tables, figures, page fields, page-limit estimate, and evidence annex. Category: `Uncategorized`. Source: `app/nodes/horizon_docx_renderer.py:111`.
21. **HorizonEvaluationAgent** — Score Excellence, Impact, and Implementation with independent cross-provider evaluators plus deterministic evidence gates. Category: `Uncategorized`. Source: `app/nodes/horizon_evaluation.py:42`.

22. **HorizonHTMLProposalRenderer** — Convert proposal HTML or Markdown into a citation-aware Horizon Europe Part B PDF with cover, TOC, page checks, and evidence annex. Category: `Uncategorized`. Source: `app/nodes/html_proposal_renderer.py:98`.
23. **HumanInLoopAgent** — Pause for human approval, rejection, or edit. Category: `Uncategorized`. Source: `app/nodes/human_in_loop.py:157`.
24. **IntegrationAgent** — Browse and pull files from a connected Google Drive or OneDrive account — provider differences live in adapters. Category: `Uncategorized`. Source: `app/nodes/integration.py:206`.
25. **InternalProjectEvidenceRetrieverAgent** — Retrieve partner, pilot, work-plan, budget and approved internal database facts; require an exact source passage and explicit human approval before drafting. Category: `Uncategorized`. Source: `app/nodes/internal_project_evidence_retriever.py:126`.
26. **KimiVisionAgent** — Analyse an uploaded image with Kimi K3. Category: `Uncategorized`. Source: `app/nodes/kimi_vision_agent.py:70`.
27. **KnowledgeRetrieval** — Retrieve secured knowledge through a saved Retrieval Profile without generating an answer. Category: `Uncategorized`. Source: `app/nodes/knowledge_retrieval.py:90`.
28. **Literal** — Emits a literal config value as its output. Category: `Uncategorized`. Source: `app/nodes/_stubs.py:33`.
29. **MCPAgent** — LLM-driven agent loop using MCP tools. Category: `Uncategorized`. Source: `app/nodes/mcp_agent.py:86`.
30. **MCPToolAgent** — Call one capability on a connected MCP server — CRM, ERP, or any other business system. Category: `Uncategorized`. Source: `app/nodes/mcp_tool.py:154`.
31. **MethodologyEngineeringAgent** — Produce one skill-guided Method Card per frozen-concept objective: method, baseline, validation, uncertainty handling, and failure condition, grounded only in verified claims and known research questions. Category: `Uncategorized`. Source: `app/nodes/methodology_engineering.py:129`.
32. **MinIOEvidenceIngestion** — Index acquired full-text sources (from MinIO pages.json) into Weaviate, stamping each chunk with its global citation display_number so retrieved passages carry their footnote number to the drafter. Category: `Uncategorized`. Source: `app/nodes/minio_evidence_ingestion.py:108`.
33. **OpenAIImageGenerationAgent** — Generate an image with an approved OpenAI image model and store the bytes in object storage. Category: `Uncategorized`. Source: `app/nodes/openai_image_generation_agent.py:89`.
34. **PaperQAEvidenceSynthesizerAgent** — Run PaperQA2 over already-acquired full-text documents, per claim, for per-document coverage and gap-aware literature synthesis. Category: `Uncategorized`. Source: `app/nodes/paperqa_evidence_synthesizer.py:131`.
35. **PDFProposalRenderer** — Render proposal sections to a styled PDF (corporate, professional, warm). Category: `Uncategorized`. Source: `app/nodes/pdf_tool.py:132`.
36. **PDFTextExtractor** — Extract text from a .pdf in object storage. Category: `Uncategorized`. Source: `app/nodes/pdf_tool.py:50`.
37. **PowerPointProposalSlides** — Build a .pptx deck from proposal sections. Category: `Uncategorized`. Source: `app/nodes/powerpoint_tool.py:49`.
38. **PriorProjectRetrieverAgent** — Search official CORDIS, LIFE and EIP-AGRI project records for precedents and synergies. Category: `Uncategorized`. Source: `app/nodes/prior_project_retriever.py:139`.
39. **ProposalEvidenceFactoryAgent** — Verify proposal claims against immutable full-text pages, build an auditable citation registry, and fail closed on evidence gaps. Category: `Uncategorized`. Source: `app/nodes/proposal_evidence_factory.py:351`.
40. **ProposalSubmissionGate** — Deterministically decide whether a proposal is ready for final human approval or autonomous document export. Category: `Uncategorized`. Source: `app/nodes/proposal_submission_gate.py:108`.
41. **ProposalTruthGraphAgent** — Freeze a drafting-safe truth graph containing only verified or qualified evidence links, plus explicit gaps and approval items. Category: `Uncategorized`. Source: `app/nodes/proposal_truth_graph.py:149`.
42. **PythonSnippetAgent** — Run a short Python snippet as a workflow step, in an isolated sandbox with no network access and no access to this platform's own code or secrets. Category: `Uncategorized`. Source: `app/nodes/python_snippet.py:100`.
43. **RAGAgent** — Hybrid retrieval + grounded answer with citations. Category: `Uncategorized`. Source: `app/nodes/rag.py:156`.
44. **ResearchSourceAcquirer** — Resolve and store bounded Deep Research citations as immutable HTML or PDF source versions for exact-passage verification. Category: `Uncategorized`. Source: `app/nodes/research_source_acquirer.py:100`.
45. **RouterAgent** — Branch the workflow on a field value, on business conditions, or via model judgment — with the reason recorded. Category: `Uncategorized`. Source: `app/nodes/router.py:254`.

- 46. **ScientificResearchPlannerAgent** — Turn the call, selected concept, and proposal graph into several bounded research briefs, each routed through approved K-Dense skills. Category: `Uncategorized`. Source: `app/nodes/scientific_research_planner.py:134`.
- 47. **ScientificSkillAgent** — Scientific synthesis or proposal work guided by approved K-Dense Agent Skills. Category: `Uncategorized`. Source: `app/nodes/scientific_skill_agent.py:67`.
- 48. **SQLQueryAgent** — Run a read-only SQL query against the business-records database, for a lookup the classified MCP tools don't already cover. Category: `Uncategorized`. Source: `app/nodes/sql_query.py:108`.
- 49. **StartAgent** — How this workflow begins: a business-friendly input form, or a conversational chatbot entry point. Category: `Uncategorized`. Source: `app/nodes/start.py:117`.
- 50. **StructuredDatasetRetrieverAgent** — Retrieve bounded Eurostat/official structured data with explicit filters, immutable snapshots, row hashes, count reconciliation, and auditable provenance. Category: `Uncategorized`. Source: `app/nodes/structured_dataset_retriever.py:139`.
- 51. **SubprocessAgent** — Run another saved workflow as a reusable business subprocess. Category: `Uncategorized`. Source: `app/nodes/subprocess.py:115`.
- 52. **TextAssemblerAgent** — Join: waits for multiple upstream branches (e.g. Category: `Uncategorized`. Source: `app/nodes/text_assembler.py:64`.
- 53. **TransformAgent** — Pure LLM transform: summarize, classify, rewrite, extract. Category: `Uncategorized`. Source: `app/nodes/transform.py:301`.
- 54. **WebSearchAgent** — Search the live public web with Auto, Tavily, OpenAI, or Kimi K3. Category: `Uncategorized`. Source: `app/nodes/web_search_agent.py:85`.
- 55. **WorkflowFileLoader** — Load uploaded workflow files. Category: `Uncategorized`. Source: `app/nodes/workflow_file_loader.py:82`.
- 56. **WorkflowInputAgent** — Information entering the workflow: manual, API, document, email or a previous workflow's output, with a declared shape. Category: `Uncategorized`. Source: `app/nodes/workflow_input.py:264`.

20. Explicitly absent, retired, or deferred capabilities

These items should **not** be described as current Eurskem AI features:

| Capability | Current state | |---|---| | Multi-workflow Pipeline product/runtime | Retired from the current application and route surface | | Business View product surface | Retired from the current application | | Scheduled/time-based workflow triggers | Not registered | | Generic webhook trigger node | Not registered | | Generic Wait/Delay node | Not registered | | Generic For Each/Loop node | Deferred; no first-class registered node | | Generic Merge node | Not registered; graph fan-in and [TextAssemblerAgent](#) cover current needs | | Generic scheduled automation service | Not present | | WebSocket run streaming | Not used; authenticated SSE is the current mechanism | | Redux/Zustand/React Query state layer | Not used; state is local/context/custom-hook based | | Alertmanager/pager routing | Not currently configured in the production stack | | Formal GDPR compliance | Not claimed; the platform provides technical controls, not organizational/legal compliance |

Maintenance

Update this catalog when any of the following changes:

1. A top-level UI mode or Studio route is added or removed.
2. A FastAPI router or major resource lifecycle is added or removed.
3. A node type is registered, hidden, deprecated, or deleted.
4. A Chat experience preset or planner route is changed.
5. An integration/provider is added or retired.
6. A major runtime, security, deployment, or observability capability changes.

For lower-level code ownership and request mapping, see:

- [docs/pdf/05_FRONTEND_BACKEND_MAPPING.pdf](#)
- [docs/pdf/06_WORKFLOW_ENGINE_AND_NODE_TYPES.pdf](#)
- [docs/pdf/12_API_REFERENCE_AND_REQUEST_FLOWS.pdf](#)
- [docs/pdf/17_FILE_BY_FILE_CODE_REFERENCE.pdf](#)
- [docs/pdf/22_FEATURE_TO_CODE_MAP.pdf](#)
- [docs/pdf/23_MASTER_INDEX.pdf](#)